Practical Hands-On Questions on MAC address, VLAN, inter-VLAN routing, EtherChannel, and Spanning Tree:

MAC Address

1. How do you view the MAC address table on a Cisco switch?

```
SW1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.4327.3e01    DYNAMIC     Fa0/2
```

2. What is the purpose of a MAC address in a network?

The MAC address serves the following purposes:

- Physical Identification: Uniquely identifies devices (computers, smartphones, routers) on a local network.
- Local Data Routing: Enables network hardware, such as switches and routers, to deliver data frames specifically to the intended destination device on the same network segment.
- Hardware-Based Addressing: Unlike IP addresses, which can change, MAC addresses are permanently assigned to the hardware, allowing for consistent, low-level identification.
- Network Access Control: Network administrators use MAC addresses for security, such as allowing or denying specific devices access to a network (MAC filtering).
- Error Detection/Management: Helps with diagnostics and monitoring device traffic within a network

3. How do you configure a static MAC address on a Cisco switch?

```
Switch(config)#do show mac add
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a18.1e01    DYNAMIC     Fa0/2
  10    0011.2233.4455    STATIC      Fa0/3
Switch(config)#
Switch(config)#mac-address-table static 0011.2233.4455 vlan 10 interface fa0/3
```

VLAN

4. How do you create a VLAN on a Cisco switch?

```
Switch(config)#mac-address-table static 0011.2233.4455 vlan 10 interface fa0/3
%The Vlan does not exist
Switch(config)#vlan 10
Switch(config-vlan)#na
Switch(config-vlan)#name Extra
Switch(config-vlan)#exit
```

5. What is the purpose of VLANs in a network?

The primary purpose of a VLAN is to logically segment a single physical network into multiple, isolated virtual networks. This allows network administrators to group devices based on functional requirements (like departments) rather than their physical location, all while using the same physical hardware.

6. How do you assign a port to a VLAN on a Cisco switch?

```
Switch(config-if)#switchport access vlan 10
Switch(config-if)#do show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Extra                            active    Fa0/17
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Inter-VLAN Routing

7. How do you configure inter-VLAN routing on a Cisco router?

Sub-interface configuration with gateway IP and dot1Q VLAN assignment

```
R1#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up
up
GigabitEthernet0/0.10  10.22.30.1      YES manual up
up
GigabitEthernet0/0.20  10.22.31.1      YES manual up
```

There is need for proper interface configuration on the switch interfaces to tag access ports and permit trunk for multiple VLAN-tagged traffic to successfully get to the Router interface

```
SW1(config-if)#do show run | section interface
interface FastEthernet0/1
 switchport access vlan 20
 switchport mode access
interface FastEthernet0/2
interface FastEthernet0/3
 switchport mode trunk
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
```

8. What is the purpose of inter-VLAN routing?

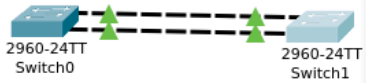
The purpose of inter-VLAN routing is to enable communication between devices on different VLANs, which are otherwise isolated from each other by design

9. How do you troubleshoot inter-VLAN routing issues?

a. Ensure/verify proper IP configurations on End User device including proper subnet mask and default gateway IP address
b. On the switch: check to ensure the proper VLAN(s) and switchport modes; with access mode assigned on end user interfaces with the appropriate VLAN assignment, and trunk mode assigned on the Router-facing interface
c. Also on the switch ensure there is no unwanted VLAN pruning on the trunk port that could potentially affect the passage of tagged traffic.
d. On the Router: ensure there is sub-interface dot1Q encapsulation that accepts tagged traffic, and proper IP configuration to ensure alignment with the intended end-user

EtherChannel

10. How do you configure EtherChannel on a Cisco switch?




```
SW2(config)#interface port-channel 01
SW2(config-if)#interface range f0/1-2
SW2(config-if-range)#channel-group 01 mode on
SW2(config-if-range)#
%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW2(config-if-range)#
SW2(config-if-range)#do show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           -      Fa0/1(P) Fa0/2(P)
SW2(config-if-range)#
```

11. What is the purpose of EtherChannel?
The primary purpose of EtherChannel is to bundle multiple physical Ethernet links into a single logical link to increase bandwidth and provide redundancy between network devices like switches, routers, and servers

12. How do you verify EtherChannel configuration on a Cisco switch?
"*show etherchannel summary*" and "*show etherchannel port-channel*" commands

```
SW2#show etherchannel port-channel
                Channel-group listing:
                ----------------------

Group: 1
----------
                Port-channels in the group:
                ---------------------------

Port-channel: Po1
------------

Age of the Port-channel   = 00d:00h:05m:04s
Logical slot/port   = 2/1       Number of ports = 2
GC                  = 0x00000000      HotStandBy port = null
Port state          = Port-channel
Protocol            =   PAGP
Port Security       = Disabled

Ports in the Port-channel:

Index   Load   Port     EC state        No of bits
------+------+------+------------------+-----------
  0     00     Fa0/1    On                 0
  0     00     Fa0/2    On                 0
Time since last port bundled:    00d:00h:04m:42s    Fa0/2
SW2#
SW2#show etherchannel summ
SW2#show etherchannel summary
```

Spanning Tree
13. How do you configure Spanning Tree on a Cisco switch?

```
SW1#
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#spanning-tree mode rapid-pvst
```

14. What is the purpose of Spanning Tree Protocol (STP)?
The primary purpose of Spanning Tree Protocol (STP) is to prevent Layer 2 loops in Ethernet networks that have redundant physical paths. By ensuring a loop-free logical topology, STP protects the network from critical failures while still allowing for backup links

15. How do you troubleshoot Spanning Tree issues on a Cisco switch?
Signs of STP failure include

a. Widespread connectivity loss to many users across a layer 2 domain,
b. MAC address flapping
c. Duplicate IP messages due to looping frames

Run one or more verification commands to map the current state of the network topology and confirm if there is an issue caused by STP

a. *show spanning-tree vlan [id]* - to identify Root Bridge and role (Root/Designated/Altn) and state (FWD/BLK) of each port
b. *show spanning-tree summary* - to get quick summary of the STP mode (PVST+, Rapid-PVST, MST and active logical ports
c. *show spanning-tree detail* - to display Topology Change Notifications (TCNs) since a rapidly increasing TCN count indicates an unstable network
d. *show processes cpu sorted* - to check if the switch's processor is too overloaded to handle BPDU traffic.

After confirming the occurrence of an STP issue, proceed to investigate the common causes

a. Duplex mismatch - these cause collisions that drop BPDUs, leading to a loop
b. PortFast Errors - if enabled on a switch-to-switch link can create transient loops

Mixed Questions

16. How do you configure a trunk port on a Cisco switch to allow multiple VLANs?

```
SW1(config)#do show cdp neigh
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
SW2          Fas 0/2          141            S       2960        Fas 0/1
SW2          Fas 0/3          141            S       2960        Fas 0/2
SW1(config)#int range f0/2-3
SW1(config-if-range)#switch
SW1(config-if-range)#switchport mo
SW1(config-if-range)#switchport mode tru
SW1(config-if-range)#switchport mode trunk
SW1(config-if-range)#switc
SW1(config-if-range)#switchport trun
SW1(config-if-range)#switchport trunk al
SW1(config-if-range)#switchport trunk allowed vl
SW1(config-if-range)#switchport trunk allowed vlan 10-30
SW1(config-if-range)#do write
Building configuration...
[OK]
SW1(config-if-range)#
SW1(config-if-range)#do show int trun
Port        Mode         Encapsulation  Status        Native vlan
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/2       10-30
Fa0/3       10-30

Port        Vlans allowed and active in management domain
Fa0/2       none
Fa0/3       none

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/2       none
Fa0/3       none
```

17. What is the difference between VLAN and inter-VLAN routing?

VLANs enables the segmenting of a physical Network into logical sub-groupings on Layer 2 while inter-VLAN routing enables communication flow between these logical groupings using Layer 3 with the functionality of a Router

18. How do you verify MAC address learning on a Cisco switch?

The *show mac address-table* command run in privileged EXEC mode calls the CAM table which contains the mapped learned MAC addresses to their specific VLAN and physical port

19. How do you configure PortFast on a Cisco switch?

```
SW1(config-if)#do show run | sec interface
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
```

Actual command *spanning-tree portfast*

```
SW1(config-if)#spanning-tree po
SW1(config-if)#spanning-tree portfast ?
  disable  Disable portfast for this interface
  trunk    Enable portfast on the interface even in trunk mode
  <cr>
SW1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
SW1(config-if)#
```

20. What is the purpose of the spanning-tree vlan command on a Cisco switch?

The *spanning-tree vlan* command is used to configure and manage Spanning Tree Protocol (STP) settings for a specific VLAN or range of VLANs which allows you tune and optimize traffic flow for your network topology as it applies to each VLAN in order to prevent Layer 2 loops